FEDERAL SPECIFICATION

VALVE, BUTTERFLY (THREADED ENDS) BRASS OR BRONZE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers class 175 screw end type butterfly valves.

1.2 Classification.

1.2.1 Class and size. Butterfly valves furnished under this specification shall be of the following types and sizes as specified (see 6.2).

Type A - Temperature range: minus 29°C (minus 20°F) to 52°C (125°F).

Type B - Temperature range: minus 34.4°C (minus 30°F) to 177°C (350°F).

Size, inch: 1/2, 3/4, 1, 1-1/4, 1-1/2, 2

1.2.2 Definitive part number. Valves shall be identified by a definitive part number (DPN). This part number is intended for cataloging and ordering purposes; it is not intended for surface printing on the valve. The DPN shall consist of the Federal Specification symbol plus the specification part number (SPN) listed in table I. The DPN shall be written as indicated below (see 6.2).

TABLE I. Available size and SPN indicator

Size (inch)	SPN 1/ Type A	Type B
1/2	104	204
3/4	106	206
1	108	208
1-1/4	110	210
1-1/2	112	212
2	116	216

1/ Last 2 digits indicate size in 1/8 inch increments.

DPN----------WWV1967 - XXX
Fed. Spec. Symbol----------|
SPN----------|

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specification:

QQ-P-416 - Plating, Cadmium, (Electrodeposited).

Federal Standard:

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-R-83428 - Rubber, Fluorocarbon Elastomer, High Temperature, Fluid, and Compression Set Resistant.
MIL-V-3 - Valves, Fittings, and Flanges (Except for Systems Indicated Herein); Packaging of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

Underwriters' Laboratories, Inc., (UL) Standard:

No. 842 - Valves for Flammable Fluids.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611; 333 Pfingston Road, Northbrook, IL 60062; 1285 Walt Whitman Road, Melville, L.I. NY 11746; 1655 Scot Boulevard, Santa Clara, CA 95050.)

3. REQUIREMENTS

3.1 Description. Butterfly valves shall be of the screwed-end design. The valve shall consist essentially of a body, disc assembly, stem, stem packing and removable packing retainer, and operating lever.

3.2 Standard product. Except for any changes necessary to comply with the requirements of this specification, the valves shall be the manufacturer's current standard commercial product.

3.3 Standard compliance.

3.3.1 Safety standard. Valves shall conform to the applicable requirements of UL Standard 842 and as hereinafter specified.

3.3.2 Certification. The supplier shall submit to the contracting officer or his authorized representative satisfactory evidence that the valves he proposes to furnish under this specification conform to the applicable requirements of UL 842. The label, or listing with re-examination, of the Underwriters' Laboratories, Inc. may be accepted as evidence that the valve conforms to this requirement. In lieu of the label, or listing with re-examination, the supplier may submit independent proof, satisfactory to the contracting officer or his authorized representative, that the valves conform to the applicable requirements of the published standards, including methods of tests, of the Underwriters' Laboratories Standard 842.

3.4 Interchangeability. All valves of the same size furnished under a specific contract shall be identical to the extent required to insure interchangeability of valves once the required spacing in piping systems is established.

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3.5 Materials. Materials for valves shall be of the type normally used for the manufacture of valves having the same body material and pressure rating as specified herein. Materials which are of dissimilar metal, with respect to the galvanic scale, shall not be used unless separated from the body by an insulating material which will avoid electrolytic corrosion. Materials shall be free from defects or imperfections that would contribute to failure of the valve or would materially reduce the usability of the valve.

3.6 Valve components.

3.6.1 Body. The valve body shall be fabricated of copper alloy conforming to the requirements of UL 842. The body shall be equipped with travel stops. The stops shall be so located that the operating lever engages the stops in the fully open and closed positions. The ends of the valve shall have hexagon wrench grips. The end shall have internal pipe threads. Valve body dimensions shall be as specified in table II. The base diameter of the valve shall be not less than the nominal pipe size diameter.

TABLE II. Valve dimensions, inches (minimum)

Size	End to end	Width between flats	Lever length stem center line to level end	Handle ht. above bottom flat
1/2	1-7/8	1-3/16	2-3/4	2-1/8
3/4	2	1-3/8	2-3/4	2-3/8
1	2-1/8	1-9/16	2-3/4	2-5/8
1-1/4	2-5/8	1-15/16	3-1/2	3-1/4
1-1/2	2-7/8	2-3/16	3-1/2	3-9/16
2	3-1/4	2-3/4	3-1/2	4-3/16

3.6.2 Disc O-ring.

3.6.2.1 Type A valve. The O-ring for type A valves shall conform to the requirements of UL 842.

3.6.2.2 Type B valve. The O-ring for type B valves shall be of a fluorocarbon elastomer. The O-ring shall conform to the requirements of MIL-R-83248 type I, class 2, O-rings (see 4.5.5).

3.6.3 Operating lever. The operating lever shall be of stainless steel or cadmium plated carbon steel. The plating shall be in accordance with QQ-P-416, class 2, type I. The lever shall engage the travel stops on the valve body. Operating levers shall be attached to the valve stem in such a manner that they can be easily replaced without the use of special tools. When the valve is open, lever handles shall be parallel to the longitudinal axis of the valve. Dimensions shall be in accordance with table II.

3.7 Pressure-temperature rating.

3.7.1 Type A valves. Type A valves shall be rated for class 175 (175 pounds per square inch gage) water, oil, gas (WOG) at any temperature in the range of minus 29°C (-20°F) to 52°C (125°F) continuous service (see 4.5.2.1 and 4.5.4).

3.7.2 Type B valves.

3.7.2.1 Class rating. Valves shall be rated for class 175 (175 pounds per square inch gage) water, oil, gas (WOG) (see 4.5.4).

3.7.2.2 Temperature rating. Valve shall be capable of continuous service at any temperature in the range of minus 34.4°C (-30°F) to 177°C (350°F) when tested in accordance with 4.5.2.2).

3.8 Marking. Each valve shall have the name or initials of the manufacturer or trade mark which can be readily identified, the size, and design working pressure distinctly cast on the body. Such marks shall not reduce the thickness of the castings by more than 0.02 inch.

3.9 Finish. All surfaces of the valve shall have an as-cast or machined finish normally produced by recognized commercial manufacturing processes. The finish shall be of the techniques established as standard practice by the valve and fittings industry.

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3.10 Workmanship. Workmanship shall be consistent with the level of quality established by the valve and fittings industry for valves produced for commercial distribution. Castings shall be free from cracks, hot tears, blowholes, porosity, or other defects affecting structural soundness. Castings which must be plugged, impregnated, brazed, or burned-in to correct defects will not be accepted. Inside and outside of castings shall be clean. Machined parts shall be free of cracks or other defects which will interfere with the proper functioning of the valve.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for lot acceptance.

4.2.1 Inspection lot. A lot shall consist of all valves of the same DPN offered for delivery at one time.

4.2.2 Sampling for inspection. Select sample size for examination in accordance with level S-4 in MIL-STD-105. Select sample size for tests in accordance with level S-2 in MIL-STD-105.

4.3 Examination and tests.

4.3.1 Examination. Examine valves for defects listed in table III. The Acceptable Quality Levels (AQL's) for defects shall be 2.5 percent defective for major defects and 6.5 percent defective for total defects.

TABLE III. Classification of defects, end item

Defect	Major	Minor
DPN (class and size) not as specified	X	
Dimensions not as specified	X	
Materials not as specified	X	
Operating lever not easily replaceable		X
Marking not as specified		X
Threads not as specified	X	
Finish not as specified		X
Defects affecting serviceability	X	
Defects not affecting serviceability		X

4.3.2 Tests.

4.3.2.1 AQL. Test sample selected in accordance with 4.2.2 as specified in 4.5.4. The AQL for this test shall be 1.0 percent defective.

4.3.2.2 Frequency of tests. The tests described in 4.5.1, 4.5.2, 4.5.3, and 4.5.5 shall be conducted on one valve from the first lot produced for each size. The tests shall be performed thereafter only upon change of manufacturing facilities, material, or design.

4.4 Certification (type B O-rings only). In lieu of performing the tests referenced in 4.5.5, the supplier or his responsible official may certify in writing to the contracting officer that the valve disc O-rings conform to the requirements of this specification.

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4.5 Test methods.

4.5.1 Deformation test. Test in accordance with UL 842.

4.5.2 Endurance test.

4.5.2.1 Type A valve. Test in accordance with UL 842.

4.5.2.2 Type B valve. Test in accordance with UL 842 except that the temperature of the test shall be as specified in 3.7.2.2 and the test media shall be dry air.

4.5.3 Mercurous-nitrate immersion test. Test in accordance with UL 842.

4.5.4 Hydrostatic strength test. Test in accordance with UL 842.

4.5.5 O-ring test, type B valve. Test type B valve O-rings in accordance with MIL-R-83248 (see 3.6.2.2).

4.6 Inspection of preparation for delivery requirements. An inspection shall be made to determine whether the packaging, packing and marking comply with the requirements of MIL-V-3.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. The valves shall be preserved, packaged and packed in accordance with MIL-V-3. The level of preservation and packaging shall be A or C and the level of packing shall be A, B or C, as specified (see 6.2).

5.2 Marking. Marking shall be in accordance with 5.2.1 or 5.2.2 as specified (see 6.2).

5.2.1 Civil agencies. Marking for shipment shall be in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. Marking for shipment shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Valves covered by this specification are intended for general use in piping systems conveying water, oil or gas.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) DPN (type and size, see 1.2.2).
(c) Level of preservation, packaging and packing (see 5.1).
(d) Whether civil or military marking is required (see 5.2).

MILITARY INTEREST:

Custodian:

Army - ME

MILITARY COORDINATING ACTIVITY:

Army - ME

CIVIL AGENCY COORDINATING ACTIVITIES:

DOT-ACO
GSA-FSS
NASA-JFK

PREPARING ACTIVITY:

GSA-FSS

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